Kaiyuan Liu

Seattle lky04@cs.washington.edu kaiyuanliu04.github.io Scholar Kaiyuan Liu

Education

B.S./M.S. **University of Washington, Seattle** 2022 ~2027
Combined B.S./M.S. Program in Computer Science
Double Major: Computer Science, Mathematics

- Mentored by Luke Zettlemoyer & Natasha Jaques
- GPA: 3.93/4.0; Honor Math Sequence; Dean's List
- Departmental Honors in Computer Science

Awards

Best Senior Thesis Award, Paul G. Allen School of Computer Science & Engineering 2026.06
ICPC World Final Honor, ICPC Foundation 2024.09
CRA Undergraduate Research Award Honorable Mention, CRA 2025.09
Shenoy Undergraduate Research Fellowship, Simons Foundation 2024 ~2025
CSE Award for Excellence Scholarship, University of Washington 2024 ~2025
ICPC North American Championship 12th Place, ICPC Foundation 2024.05
UW Winter Programming Contest Champion, University of Washington 2023 ~2024
Tinker Research Grant, Thinking Machines Lab 2025.11

Publications

When Agents Slow Down: Understanding LLM Agents' Test-Time Strategies via Elo-per-token Analysis
Kaiyuan Liu, Qiuyang Mang, Bo Peng, Wenhao Chai, Hanchen Li, Shreyas Pimpalgaonkar, Luke Zettlemoyer, Alex Dimakis, Alvin Cheung
arxiv.org/abs/2609.15309
arXiv Preprint, 2026

AutoCode-RL: Climbing the Difficulty Pyramid with Verifiable Synthetic Problems
Kaiyuan Liu, Zhiming Wang, Zeyu Shen, *et al.*
OpenReview Link
ICLR 2027 Under Review

AutoCode: LLMs as Problem Setters for Competitive Programming
Shang Zhou*, Zihan Zheng*, **Kaiyuan Liu***, *et al.* *Equal contribution
arxiv.org/abs/2510.12803
The Fourteenth International Conference on Learning Representations (ICLR 2026)

LiveCodeBench Pro: How Do Olympiad Medalists Judge LLMs in Competitive Programming?
Zihan Zheng*, Zerui Cheng*, Zeyu Shen*, Shang Zhou*, **Kaiyuan Liu***, *et al.* *Equal contribution
arxiv.org/abs/2506.11928
The Thirty-Ninth Annual Conference on Neural Information Processing Systems (NeurIPS 2025)

Harbor Adapters and Harbor-Index: Infrastructure and a Curated Meta-Dataset for Large-Scale Agentic Evaluation
Lin Shi, Haowei Lin, Zixuan Zhu, …, **Kaiyuan Liu**, *et al.*
arxiv.org/abs/2609.04298
The Fortieth Annual Conference on Neural Information Processing Systems (NeurIPS 2026)

Agents' Last Exam

Yiyou Sun, Xinyang Han, Weichen Zhang, …, **Kaiyuan Liu**, …, Dawn Song

arxiv.org/abs/2606.05405
The Fortieth Annual Conference on Neural Information Processing Systems (NeurIPS 2026)

FrontierCS 2: A Benchmark for Computer Science R&D

Qiuyang Mang, Wenhao Chai, Runyuan He, …, **Kaiyuan Liu**, *et al.*

arxiv.org/abs/2610.29932
arXiv Preprint, 2026

EmbodiedSWE: Coding Agents for Long Horizon Dexterous Robotics

Zeyu Shen, Haoxiang You, Yilang Liu, …, **Kaiyuan Liu**, *et al.*

arxiv.org/abs/2609.27308
arXiv Preprint, 2026

OpenDeepThink: Parallel Reasoning via Bradley-Terry Aggregation

Shang Zhou, Wenhao Chai, **Kaiyuan Liu**, Huanzhi Mao, Qiuyang Mang, Jingbo Shang

arxiv.org/abs/2605.15177
ICLR 2027 Under Review

FrontierSmith: Synthesizing Open-Ended Coding Problems at Scale

Runyuan He, Qiuyang Mang, Shang Zhou, **Kaiyuan Liu**, Hanchen Li, Huanzhi Mao, *et al.*

arxiv.org/abs/2605.14445
The Fortieth Annual Conference on Neural Information Processing Systems (NeurIPS 2026) *Spotlight*

FrontierCS: Evolving Challenges for Evolving Intelligence

Qiuyang Mang[*], Wenhao Chai[*], Zhifei Li[*], Huanzhi Mao[*], Shang Zhou[*], …, **Kaiyuan Liu**, *et al.* [*]Equal contribution

arxiv.org/abs/2512.15699
Forty-Third International Conference on Machine Learning (ICML 2026)

Evaluating LLM Agents as Human Simulators in Climate Social Dilemmas

Kaiyuan Liu[*], Xiaoxuan Hou[*], Jiayi Yuan, Natasha Jaques [*]Equal contribution

OpenReview Link
ICLR 2026 MALGAI Workshop *Oral*

Blogpost

Humans Still Beat AI in the Long Horizon: Revisiting Test-Time Scaling in the Agent Era

Qiuyang Mang, **Kaiyuan Liu**, Bo Peng, Shreyas Pimpalgaonkar, Luke Zettlemoyer, Alex Dimakis, Alvin Cheung

frontier-cs.org/blog/humans-still-beat-ai/
Frontier-CS Blog, June 16, 2026

Your Next Long-Context Recipe: Open-Ended Problems

Qiuyang Mang, **Kaiyuan Liu**, Zhifei Li, Wenhao Chai, Hanchen Li, Lin Shi, Zhixuan Zhu, Crystal Zhou, Alex Dimakis

frontier-cs.org/blog/harbor/
Frontier-CS Blog, May 12, 2026

Experience

Microsoft AI (formerly Microsoft Research – AI Frontiers), Research Intern — Seattle, WA
2026.06 ~2026.09

- Research on self-improvement.
- Advised by Dr. Asli Celikyilmaz.

Social RL Lab, Research Assistant — Seattle, WA
2025.01 ~Present

- Working on various projects related to multi-agent, reinforcement learning, and large language models.
- Mentored by Dr. Natasha Jaques.

Allen Institute, Research Assistant — Seattle, WA
2024.09 ~2025.05

- Applied Deep Learning and Reinforcement Learning to analyze representations from artificial models and neural data from Multi-Armed Bandit mouse foraging tasks. Explored brain representations and dynamics in decision making

VecML, Machine Learning Engineer — Seattle, WA
2024.06 ~2024.09

- Tested and developed vector databases for Retrieval-Augmented Generation.
- Designed and implemented an Memory-Disk Hybrid Architecture for fast Top-K Nearest Neighborhood Search

University of Washington, Teaching Assistant — Seattle, WA
2023 ~2024

- Assisted in undergraduate algorithm courses for both major (CSE 421) and non-major (CSE 417) students.
- Served as leading TA in CSE 421 2024 Winter.

Professional Service

Reviewer, NeurIPS Workshop 2025, NeurIPS 2026, AAAI 2026, AAMAS 2026
ICPC Coach, University of Washington ICPC 2026

Technologies

Programming: Python, PyTorch, JAX, C++, Java, TypeScript, LaTeX

Databases: MySQL, NoSQL, PostgreSQL, SQLite, Azure

Topics: Algorithm Design, Reinforcement Learning, Machine Learning, NLP, Computational Neuroscience

Languages: English, Chinese (Mandarin)